

OBSERVATION TASK

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

HTML5 Graphics and CSS3 Web Styling -My Styled Webpage

2. AIM

To create “My Styled Webpage” using HTML5 Canvas and CSS3 styling properties such as colors, fonts, borders, margins, padding, gradients, shadows, rounded corners, and hover effects.

3. OBJECTIVE

- To draw a rectangle, circle, line, and text using HTML5 Canvas.
- To understand the use of JavaScript with the Canvas element.
- To apply CSS3 properties to design a webpage.
- To create a header, navigation bar, three content cards, and footer.
- To implement gradient backgrounds, borders, rounded corners, and shadows.
- To implement hover effects for navigation links and buttons.

4. THEORY

The HTML5 <canvas> element provides a rectangular area for drawing graphics directly on a webpage. Graphics are created using JavaScript and the Canvas 2D API. It can be used to draw shapes, lines, text, and other graphical elements.

In this experiment, the Canvas is used to draw:

- A **blue rectangle** using fillRect().
- A **red circle** using arc().

- A **black line** using moveTo(), lineTo(), and stroke().
- The name **REDDIPUJITHARAM** using fillText().

CSS3 provides properties for **colors, fonts, borders, margins, padding, backgrounds, gradients, shadows, rounded corners, and hover effects**. In this experiment, CSS3 is used to design the header, navigation bar, three content cards, buttons, and footer.

Thus, HTML5 Canvas and CSS3 are combined to create an attractive and styled webpage.

5. ALGORITHM / PROCEDURE

1. Create an HTML file and set the title as **“My Styled Webpage”**.
2. Create a Canvas element with a width of 500 pixels and height of 300 pixels.
3. Access the Canvas using JavaScript.
4. Obtain the 2D drawing context.
5. Draw a blue rectangle using fillRect().
6. Draw a red circle using arc().
7. Draw a black line using moveTo() and lineTo().
8. Display REDDIPUJITHARAM using fillText().
9. Create the webpage header.
10. Create a navigation bar with Home, About, Services, and Contact links.
11. Create three content cards.
12. Add a button to each card.
13. Create a footer.
14. Create a separate style.css file.
15. Apply a gradient background to the webpage.
16. Apply colors, fonts, padding, borders, border radius, and shadows.
17. Apply hover effects to navigation links and buttons.
18. Link the CSS file with the HTML file.
19. Open the webpage in a browser and verify the output.

6. PROGRAM

PART-A

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
</head>

<body>
  <h2>Canvas Example</h2>
  <canvas id="myCanvas" width="500" height="300" style="border:2px solid rgb(20, 1, 1);"></canvas>
  <script>
    let canvas = document.getElementById('myCanvas');
    let ctx = canvas.getContext("2d");
    ctx.fillStyle = "blue";
    ctx.fillRect(30, 30, 120, 70);

    ctx.beginPath();
    ctx.arc(250, 65, 35, 0, 2 * Math.PI);
    ctx.fillStyle = "red";
    ctx.fill();

    ctx.beginPath();
    ctx.moveTo(30, 150);
    ctx.lineTo(250, 150);
    ctx.strokeStyle = "black";
    ctx.lineWidth = 5;
    ctx.stroke();

    ctx.font = "24px Arial";
```

```
        ctx.fillStyle = "green";
        ctx.fillText("REDDIPUJITHARAM", 30, 230);
    </script>
</body>

</html>
```

PART-B

```
//.html
<!DOCTYPE html>
<html>
<head>
    <title>My Styled Webpage</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <header>
        <h1>Welcome to My Styled Webpage</h1>
    </header>
    <nav>
        <a href="#">Home</a>
        <a href="#">About</a>
        <a href="#">Services</a>
        <a href="#">Contact</a>
    </nav>
    <section class="cards">

        <div class="card">
            <h2>Card 1</h2>
            <p>This is the first content card.</p>
            <button>Learn More</button>
        </div>

        <div class="card">
            <h2>Card 2</h2>
            <p>This is the second content card.</p>
            <button>Learn More</button>
        </div>
```

```

        <div class="card">
            <h2>Card 3</h2>
            <p>This is the third content card.</p>
            <button>Learn More</button>
        </div>

</section>

<!-- Footer -->
<footer>
    <p>&copy; 2026 My Styled Webpage. All Rights Reserved.</p>
</footer>

</body>
</html>

//.css
/* Gradient Background */
body {
    font-family: Arial, sans-serif;
    background: linear-gradient(to right, lightblue, rgb(238, 197, 144));
}
header {
    background-color: darkblue;
    color: white;
    text-align: center;
    padding: 20px;
    text-shadow: 2px 2px 5px rgb(201, 22, 22);
}
nav {
    background-color: rgb(106, 198, 14);
    padding: 15px;
    text-align: center;
}
nav a:hover {
    color: yellow;
    text-decoration: underline;
}

```

```

.cards {
  display: flex;
  gap: 30px;
  margin: 30px;
}
/* Individual Card */
.card {
  background-color: white;
  width: 250px;
  padding: 2px;
  border: 2px solid navy;
  border-radius: 15px;
  box-shadow: 5px 5px 10px gray;
  text-align: center;
}
.card h2 {
  font-family: "Times New Roman", serif;
  font-size: 24px;
}
button:hover {
  background-color: orange;
}
button {
  background-color: green;
  color: white;
  padding: 10px 20px;
  border: none;
  border-radius: 8px;
  cursor: pointer;
}

```

7. CODE EXPLANATION

Code	Explanation
<canvas>	Creates the HTML5 drawing area.
getContext("2d")	Provides the 2D drawing context.

fillStyle	Sets the color used for filling shapes or text.
fillRect()	Draws a filled rectangle.
beginPath()	Starts a new drawing path.
arc()	Creates a circular arc or circle.
fill()	Fills the created shape with the selected color.
moveTo()	Sets the starting point of a line.
lineTo()	Sets the ending point of a line.
stroke()	Draws the line/path.
fillText()	Displays text on the Canvas.
<header>	Defines the header section.
<nav>	Defines the navigation bar.
.cards	Creates the container for the three cards.
.card	Styles each individual card.
linear-gradient()	Creates a gradient background.
Padding	Adds space inside an element.
Margin	Adds space outside an element.
Border	Adds a border around an element.
border-radius	Creates rounded corners.
box-shadow	Adds a shadow around cards.
text-shadow	Adds a shadow to header text.
display: flex	Arranges the cards horizontally.
:hover	Applies styling when the mouse pointer moves over an element

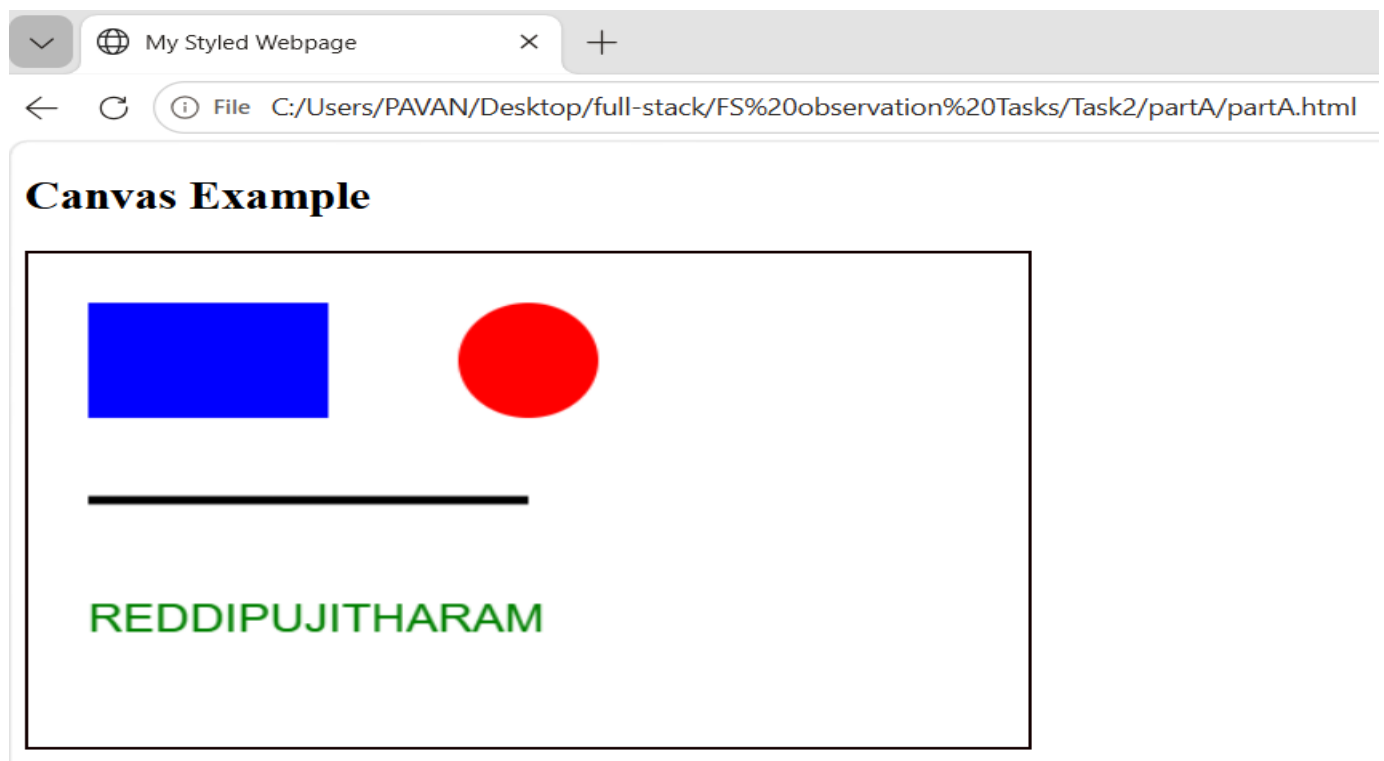
8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file	Webpage should load successfully	Webpage loaded successfully	Pass
TC-02	Check Canvas graphics	Rectangle, circle, line and name should be displayed	All graphics displayed	Pass
TC-03	Check navigation bar	Home, About, Services and Contact links should be displayed	Links displayed	Pass
TC-04	Hover over navigation links	Link color should change to yellow and	Hover effect applied	Pass

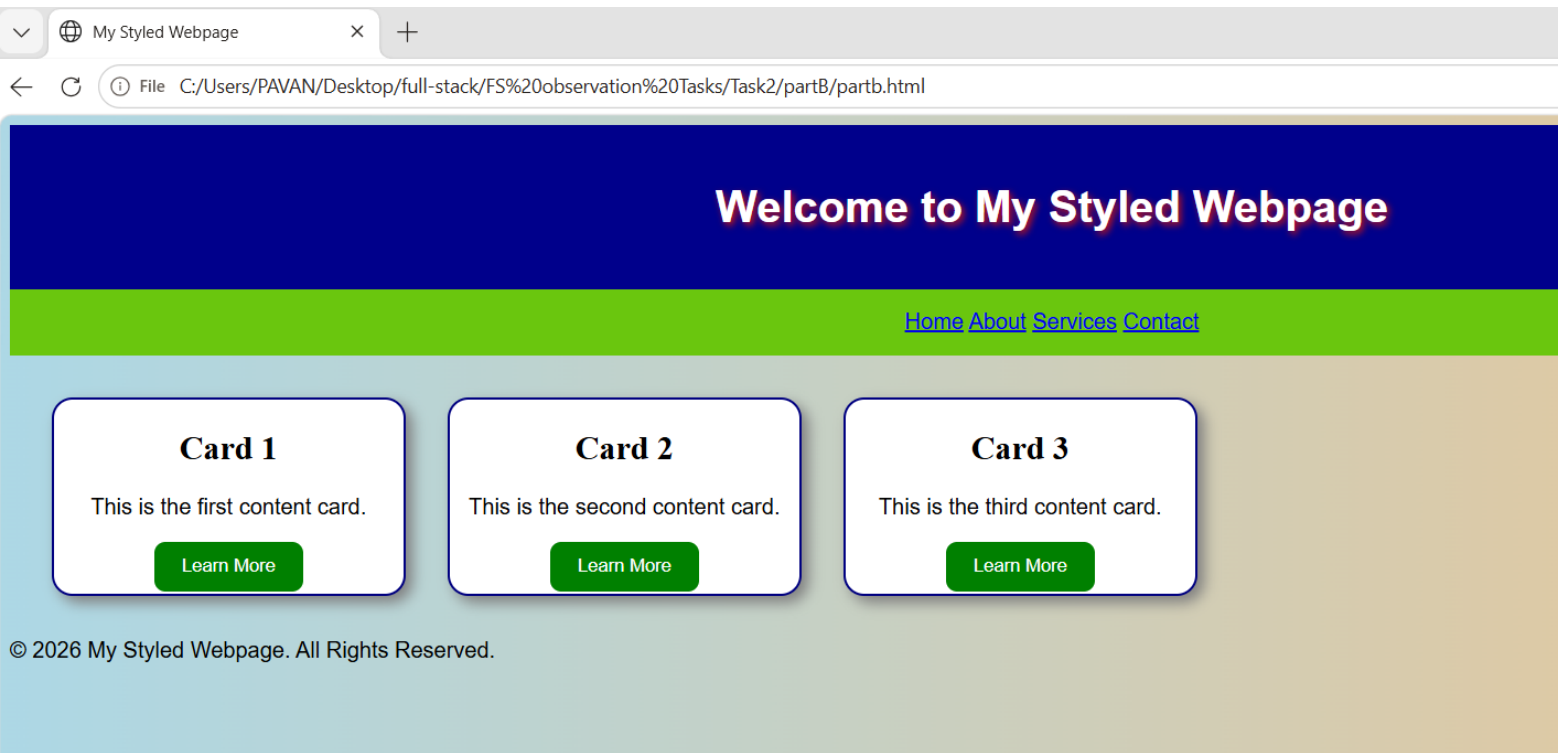
		underline should appear		
TC-05	Check three content cards	Three cards with buttons should be displayed with proper styling	Three cards displayed	Pass
TC-06	Hover over buttons	Button background color should change to orange	Button hover effect applied	Pass

9. OUTPUT

PART-A



PART-B



10. RESULT

The “My Styled Webpage” was successfully created using HTML5 Canvas and CSS3. The Canvas successfully displays a rectangle, circle, line, and name. The webpage contains a header, navigation bar, three content cards, and footer with CSS3 features such as gradient background, fonts, borders, padding, margins, box shadow, text shadow, rounded corners, and hover effects.